

* Max Subarray Sum-:

Idea-:

for every subarray of size k iterate & calculate it's sum
over all take the max sum.

* optimisation-:

- 1) Construct prefix array - $pf(N)$
- 2) Calculate max sum

Tc: $O(N)$

Sc: $O(N)$

* Optimisation-:

carry + All subarrays $\Rightarrow$ sliding
forward of same size window

$$\text{Sum} = \text{Sum} - A[s-1] + A[e]$$

Tc: $O(N)$

Sc: $O(1)$

* Min Swaps:-

→ Count of all elements $\leq B$, say k

→ size of subarray will be fixed k

All elements $\leq B \rightarrow$ good

All elements $> B \rightarrow$ bad

No of swaps in = no. of bad elements
a window in that window

TC: $O(N)$

SC: $O(1)$

* print boundary in clockwise direction:

Idea:-

1. first row $N-1 \rightarrow$

2. last column $N-1 \downarrow$

3. last row $N-1 \leftarrow$

4. first column $N-1 \uparrow$

4 loops

TC: $O(N)$

SC: $O(1)$

* Spiral form (print matrix) -:

1) Keep on printing boundary of box all you can

2) Reduce the size of box after each cycle.

$$TC: O(N^2)$$

$$SC: O(1)$$